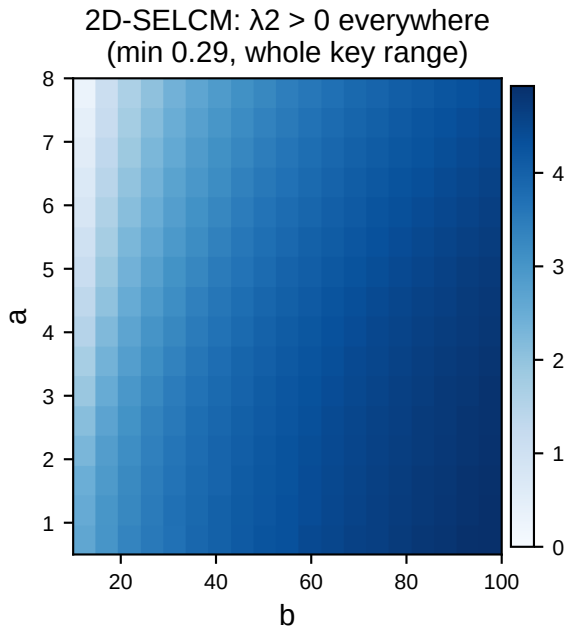
Hyperchaotic lattice audio cipher with provable trail bounds



Key + nonce → SHAKE-256

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256-lane chaotic lattice

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Keyed GF(2^{16}) S-box (DU 4)

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3-pass cyclic diffusion

≥ 13 active S-boxes
trail probability $\leq 2^{-182}$

NSCR 99.9985 %
156 files, 30 keys

CPA / KPA ≤ 0.003 %
vs 100 % for baselines

51.5 MB/s in C
292× real time